

One endomorphism, one telescope

The abstract scaffold behind DobbertinLego: Dobbertin's Theorem 1, step $(1) \Rightarrow (2)$

A picture book of the DobbertinLego Lean library, organized around a single abstract gadget — `import DobbertinLego`

The idea. The first step of Dobbertin's Theorem 1 — the implication equation $(1) \Rightarrow$ equation (2) — rests on *one* abstract object:

an abelian group A together with an endomorphism $\varphi : A \rightarrow A$.

Equivalently A is a module over $\mathbb{Z}[t]$ with t acting as φ ; the endomorphism lives in the ring $\text{End}(A)$, where multiplication is composition. Inside that ring are the two elements that generate everything:

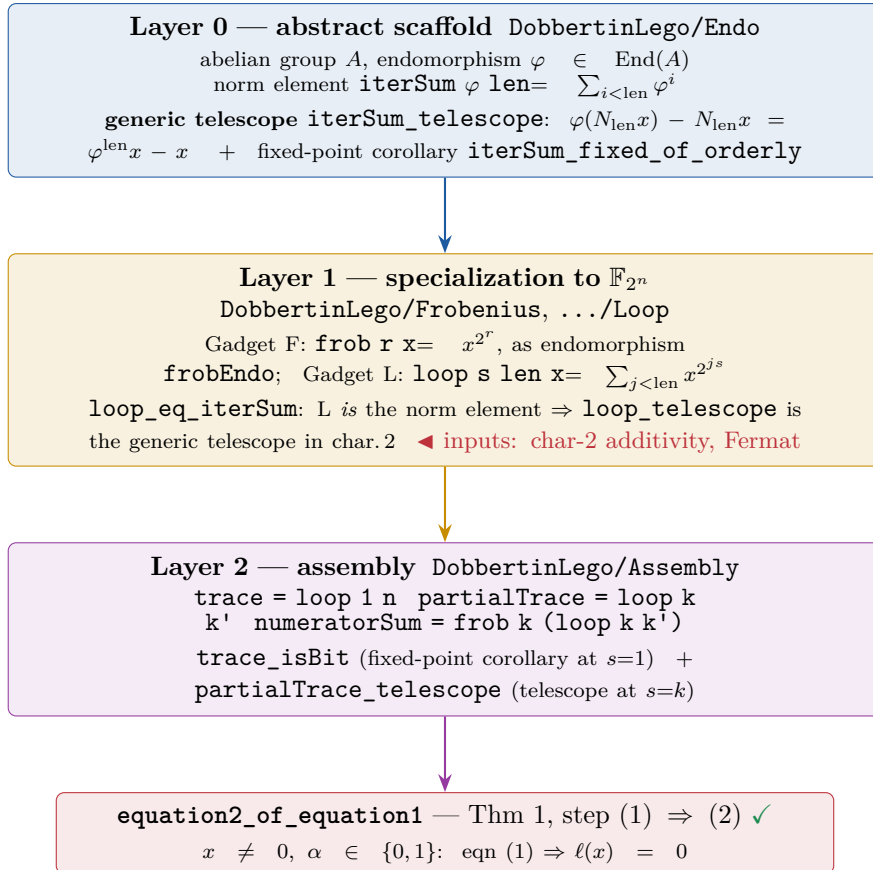
$$\text{norm / geometric element} \quad N_m = \sum_{i < m} \varphi^i, \quad \text{augmentation} \quad \wp = \varphi - 1.$$

The single load-bearing algebraic fact is the **geometric telescope**

$$\wp \circ N_m = (\varphi - 1) \sum_{i < m} \varphi^i = \varphi^m - 1$$

the additive shadow of the geometric series / additive Hilbert 90. It holds for *any* endomorphism of *any* abelian group. Both telescoping identities Dobbertin uses, and the fact that a trace is a bit, are specializations of it. Every box below is a ✓ sorry-free Lean declaration; the headline uses only `propext`, `Classical.choice`, `Quot.sound`.

Three layers on one page



The arrows are literal Lean imports. The abstract layer *organizes* the proof; the arithmetic content (characteristic 2, Fermat) is fed in exactly once, at the Layer 0→Layer 1 boundary.

1 Layer 0 — the abstract scaffold (Endo.lean)

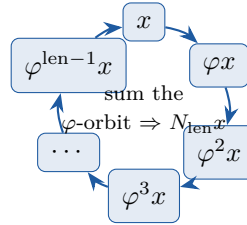
Nothing here mentions fields, Frobenius, or characteristic. The whole layer is about an endomorphism of an abelian group and its norm element.

```
variable {A : Type*} [AddCommGroup A]

def iterSum (φ : AddMonoid.End A) (len : ℕ) (x : A) : A := ∑_{i<len} (φ^i) x

iterSum_telescope      : φ(iterSum φ len x) - iterSum φ len x
                      = (φ^len) x - x                                ✓ -- additive Hilbert 90
iterSum_fixed_of_orderly : φ^n = 1 ⇒
                      φ(iterSum φ n x) = iterSum φ n x              ✓ -- norm lands in fixed points
```

The *telescope*, in one line: applying one more φ to N_{len} and subtracting N_{len} cancels every interior iterate $\varphi^1, \dots, \varphi^{\text{len}-1}$, leaving the endpoints $\varphi^{\text{len}} - 1$. The *fixed-point corollary* is the same identity when φ has finite order ($\varphi^n = 1$, so $\varphi^n - 1 = 0$): the full norm element is φ -invariant, i.e. it lands in the fixed subgroup.



2 What is a genuine input, and what is formal

The value of the abstract layer is that it makes visible *which* facts are generic module theory and *which* are irreducible arithmetic inputs of the finite field. The three “load-bearing” facts split cleanly:

fact	abstract home	status
Freshman’s dream $(a+b)^2 = a^2+b^2$	φ is an <i>additive</i> endomorphism (lives in $\text{End}(A)$)	genuine char-2 input — Mathlib frobenius, add_pow_char_pow
Telescoping $S^{2^k} + S = x^2 + x$	$(\varphi - 1) \sum \varphi^i = \varphi^m - 1$	formal — pure module theory (iterSum_telescope)
$\text{Tr}(x) \in \{0, 1\}$	norm element lands in fixed subgroup ($\varphi^n = 1$)	formal skeleton (iterSum_fixed_of_orderly) + input $\mathbb{F}_2 = \{0, 1\}$
$\varphi^n = \text{id}$ (Fermat)	φ has finite order n (cyclic Galois group)	finite-field input — Mathlib FiniteField.pow_card
$k'k \equiv 1$ collapses $\varphi^{k'k}$ to φ	multiplication in $\mathbb{Z}/n = \text{ord}(\varphi)$	formal — periodicity frob_periodic

In one sentence: category/module theory supplies the *shape* (the telescope on a dualizable / finite-order object); characteristic 2 supplies only two atoms — φ is additive, and the fixed subgroup is the two-element field.

3 Layer 1 — specialization to $\mathbb{F}_{2^n}$ (Frobenius, Loop)

The bridge is short: the Frobenius $x \mapsto x^{2^s}$ is additive (frob_add), so it *is* an element `frobEndo s` of $\text{End}(F)$, its iterates are its ring powers, and Fermat is finite order.

```

def frob (r : ℕ) (x : F) : F := x ^ (2 ^ r) -- Gadget F
def frobEndo (s : ℕ) : AddMonoid.End F := AddMonoidHom.mk' (frob s) (frob_add s)

frobEndo_pow_apply : (frobEndo s ^ j) x = frob (j*s) x      ✓ -- iterates = ring powers
frobEndo_pow_card  : |F|=2^n ⇒ frobEndo n = 1                ✓ -- Fermat = finite order

def loop (s len : ℕ) (x : F) : F := ∑_{j<len} frob (j*s) x    -- Gadget L
loop_eq_iterSum : loop s len x = iterSum (frobEndo s) len x  ✓ -- L IS the norm element

loop_telescope : frob s (loop s len x) + loop s len x
                = frob (len*s) x + x                          ✓ -- generic telescope, char 2

```

Because *loop* is the abstract norm element, *loop_telescope* is no longer a bespoke induction: it is *iterSum_telescope* read in characteristic 2, where $-1 = +1$ turns the subtraction into addition.

4 Layer 2 — assembly, and the headline

The paper's three objects are one settings of the *same* norm element, and its two auxiliary facts are two specializations of the *same* scaffold.

paper object	abstract element	Lean
trace $\text{Tr}(x)$	N_n at $\varphi = \text{Frob}^1$	<code>trace n x = loop 1 n x</code>
partial trace $P(x)$	$N_{k'}$ at $\varphi = \text{Frob}^k$	<code>partialTrace k k' x = loop k k' x</code>
numerator sum $S(x)$	$\varphi^k N_{k'}$	<code>numeratorSum k k' x = frob k (loop k k' x)</code>
$\text{Tr}(x) \in \{0, 1\}$	fixed-point corollary	<code>trace_isBit</code>
$S + P = x^2 + x$	telescope at $s=k$	<code>partialTrace_telescope</code>

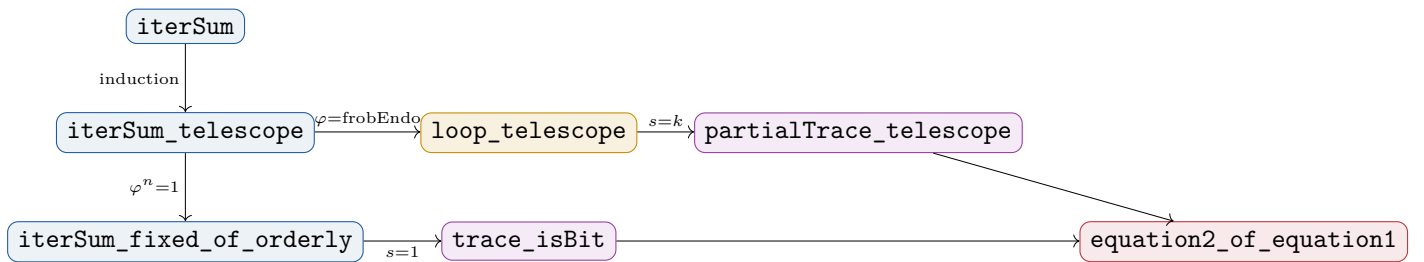
```

theorem equation2_of_equation1 (hn : |F| = 2^n) (hkk' : k*k' % n = 1)
  (hα : α=0 ∨ α=1) (hx : x ≠ 0) (h : equation1 n k k' α c x) :
  linearized k c x = 0

```

The proof collapses $\alpha \cdot \text{Tr}(x)$ to a bit (via `trace_isBit`), solves (1) for P using $S = P^{2^k}$ and $S + P = x^2 + x$, raises to the 2^k power (Frobenius additive), and divides by $x^{2^k} \neq 0$ to read off $\ell(x) = 0$. *Why* $x \neq 0$: at $x = 0$ the cleared (1) holds vacuously while $\ell(0) = 1 \neq 0$.

Compositional build diagram



One sentence. Step (1) $\Rightarrow$ (2) of Dobbertin's Theorem 1 is *one* abstract gadget — an abelian group with an endomorphism, its norm element, and the single geometric telescope $(\varphi-1) \sum \varphi^i = \varphi^m - 1$ — specialized to the Frobenius of $\mathbb{F}_{2^n}$; the paper's trace, partial trace and numerator sum are that norm element at three settings, its two auxiliary facts are the telescope and its finite-order corollary, and the arithmetic of characteristic 2 enters only as the two irreducible atoms (additivity, two-element fixed field). *Scope:* the first step only; Theorem 1 proper and the difference-set half are not part of this minimal library.

Source: `DobbertinLego/ (Endo, Frobenius, Loop, Assembly)` and `DobbertinLego.lean`. Paper: H. Dobbertin, *Kasami Power Functions, Permutation Polynomials and Cyclic Difference Sets* (1999), proof of Theorem 1, step (1) $\Rightarrow$ (2).